

Evaluating reintroduction methods for the Gulf of Saint Lawrence aster (*Symphotrichum laurentianum*) on Prince Edward Island¹

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Abstract: The Gulf of St. Lawrence aster (SLA; *Symphotrichum laurentianum* (Fernald) G.L. Nesom) is an annual plant species endemic to the Gulf of St. Lawrence region. Owing to the dynamic nature of the environment that the SLA inhabits, severe and major threats to both the aster and its habitat exist. The Committee on the Status of Endangered Wildlife in Canada listed the species as threatened in Canada in 2004. This status was assigned based on the species' limited distribution, fluctuating population size, and continued pressures on its habitat. Surveys have revealed that both site and population numbers have been further and drastically reduced on Prince Edward Island. In 2007, only one populated site of 482 individuals remained. It is possible that this species has been extirpated from Prince Edward Island. Recovery of this species on Prince Edward Island is feasible. Promising results related to seeding and the transplantation of greenhouse-grown seedlings at four in situ sites demonstrated that SLA plantlets have the potential to serve as seed stock to re-establish populations. Over the 2 years of the transplantation experiment, the pooled overall survivorship was 52.8%. Specific site manipulations that were tested may also increase the potential survivorship of the transplants and facilitate second-generation germination.

Key words: *Symphotrichum laurentianum*, recovery strategy, species-at-risk, conservation, endemic, transplant, seeding.

Résumé : L'aster du golfe St-Laurent (SLA; *Symphotrichum laurentianum* (Fernald) G.L. Nesom) constitue une espèce de plante endémique de la vallée du St-Laurent. Du à la nature dynamique de l'environnement où habite le SLA, il existe des menaces majeures pour l'aster et son habitat. On retrouve l'espèce comme menacée dans la liste du Comité canadien sur le statut des espèces indigènes menacées, depuis 2004. Ce statut lui a été attribué sur la base de la distribution limitée de l'espèce, d'une fluctuation des dimensions de la population et des pressions constantes sur son habitat. Les suivis révèlent qu'à la fois les nombres de sites et de populations se sont vus encore radicalement réduits sur l'Île-du-Prince-Édouard. En 2007, on ne retrouvait qu'une seule station comptant 482 individus. Il est possible que cette espèce ait été extirpée de l'île. Il est possible de récupérer cette espèce sur l'île. Des résultats prometteurs obtenus à partir d'ensemencements et de transplantations de plantules développées en serres sur quatre sites in situ démontrent que les plantules du SLA peuvent servir de réserve de graines pour rétablir les populations. Au cours des deux années des expériences de transplantation, on a observé une survie globale regroupée de 52,8 %. Des manipulations spécifiques aux sites des expériences peuvent également augmenter le potentiel de survie des transplants et faciliter des germinations de seconde génération.

Mots-clés : *Symphotrichum laurentianum*, stratégie de recouvrement, espèces à risque, conservation, endémie, transplant, ensemencements.

Introduction

The Gulf of St. Lawrence aster (SLA; *Symphotrichum laurentianum* (Fernald) G.L. Nesom) is an annual plant species endemic to the Magdalen Islands of Quebec, New Brunswick, and Prince Edward Island (PEI) (Houle and Haber 1990; Reynolds and Houle 2001). SLA belongs to the family Asteraceae and is found within coastal ecosystems. SLA tends to grow within sandy, brackish, or muddy soils in areas that are occasionally flooded by salt water. As a result, SLA grows within a dynamic and unpredictable intertidal habitat, mainly on the upper levels of salt marshes above the high-water mark, but SLA can also be found along margins of ponds, lagoons, and coastal dunes (Houle and Haber 1990).

Plant emergence begins in mid-June and flowering occurs from late August to early September, with seed dispersal taking place in mid-October to early November (Stewart and Lacroix 2001). The plant is typically less than 30 cm tall and can have either a branched or unbranched stem. Mature leaves are spatulate to linear-lanceolate with a smooth, fleshy surface, and average 20–60 mm in length and 2–7 mm in width (Fernald 1914). Each

individual plant has the ability to produce between 1 and over 700 inflorescences (Houle and Haber 1990). These inflorescences have leafy bracts and individual flowers have a distinct white pappus (Houle 1988) (Fig. 2A). The white pappi are the showiest part of the plant and are fully expanded just before dispersal in mid-October, making it easy to identify in the field at this particular stage of development.

Although most Asteraceae outcross as the principal method of reproduction (Houle 1988), the SLA is also self-compatible and may self-fertilize (Houle and Haber 1990). However, the only form of natural regeneration for this species is through seed dispersal and germination (Lacroix et al. 2007; Kemp and Lacroix 2004). SLA seeds can be dispersed through the air or water with the aid of a pappus. After dispersal, the seeds must overwinter before germinating under favorable conditions including temperature and light levels typical of marshes of the region. It is unknown how many years SLA seeds can remain viable in nature. Under laboratory conditions, seeds of SLA have been shown to be viable after 3 years, with an estimated viability of up to 10 years (Houle and Haber 1990).

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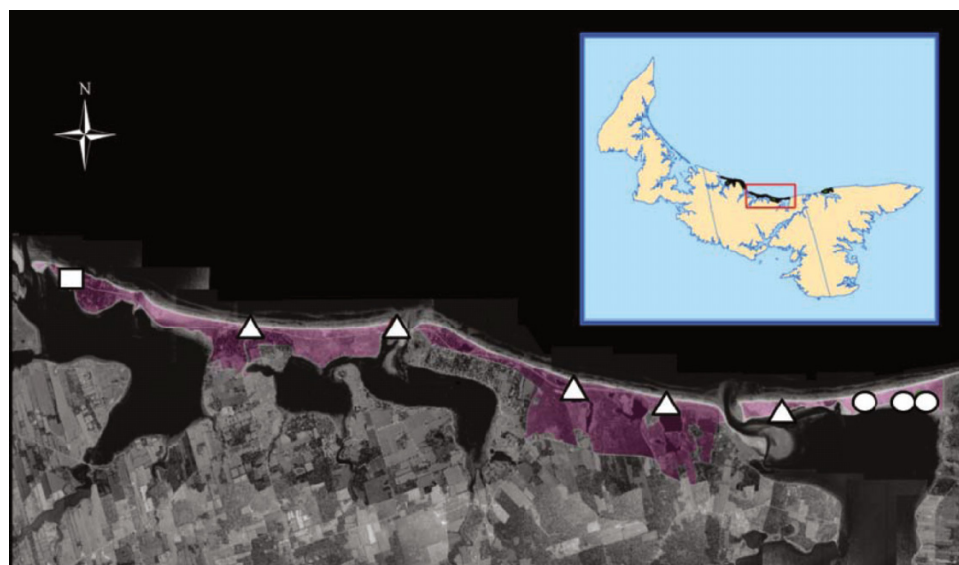
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Fig. 1. Site map for Gulf of St. Lawrence aster (SLA) in Prince Edward Island National Park. The four experimental sites, identified as a square (nonhistorical site) and three circles (historical sites), from west to east include the following: Robinson's Island, the Blooming Point Dune Slack, Blooming Point East Marsh B, and Blooming Point East Marsh A. Triangles represent historical sites.



Although SLA is currently listed as “threatened” by the Committee on the Status of Endangered Wildlife in Canada (COSEWIC) (COSEWIC 2004), long-term population monitoring on PEI indicates that the trends in both numbers of plants and sites have decreased dramatically (Jenkins 2008, K. Tulk, personal communication, 2008). Historically, eight sites were located where this species was known to exist on PEI. With the exception of one site in Tignish, all of the sites were located within Prince Edward Island National Park (PEINP). Since 1992, intermittent population surveys have been conducted at sites within PEINP by a number of researchers and organizations, e.g., individuals from the Lacroix laboratory at the University of Prince Edward Island (UPEI) (Kemp and Lacroix 2004; Allain 2007; Jenkins 2008), staff of PEINP (K. Tulk, personal communication, 2008), and staff of the Atlantic Canada Conservation Data Center (ACDC) (D. Mazerolle, personal communication, 2008). The trends depicted from the recent surveys indicate that plant numbers have declined from a high of 160 000 in seven sites in 2000 to 482 plants in one site in 2007, suggesting that the species may now be extirpated from PEI.

When present, SLA tends to be the dominant species in clearly open and relatively small patches of coastal salt marshes. SLA requires full sunlight for optimal growth and, consequently, minimal competition from surrounding vegetation for successful development (Houle 1988). Competition for light can have negative influences on this plant's development, especially during the adult stages of life both pre- and post-flowering (Houle et al. 2001). Species commonly found growing in SLA microhabitats include glasswort (*Salicornia europaea*), sea milkwort (*Glaux maritima*), and dense canopies of sedges (*Carex* spp.) (Houle and Valéry 2003; Steeves 2005).

Field observations indicate that SLA seedlings can be found growing in and around pothole-like depressions (C. Lacroix, personal communication, 2008). These depressions may be created by ice rafting (the scouring of vegetation and substrate owing to ice movement) and result in an overall decrease in vegetation density within the immediate area (Fig. 2B). The creation of these pothole patches allows for increase in light availability, water, and access to substrate, all factors which benefit SLA establishment and development (Houle and Belleau 2000; Reynolds and Houle 2001; Houle and Valéry 2003).

When it comes to plant conservation, the preservation of natural habitats is the most effective action that can be taken (Given 1994; Wilson 1992; Harrison and Bruna 1999). However, in the case of the SLA on PEI, it is evident that additional intervention may be

necessary to sustain the species. The Lacroix laboratory has been successful in producing an extant seed bank for SLA. In 1999, a bulk collection of seeds was gathered from historical locations within PEINP (Blooming Point and Covehead Pond). These seeds have been used over the years to generate an extensive bank of viable seeds that is currently stored at the UPEI. Although the plants used in this study were from a limited original population, Heard et al. (2009) demonstrated that populations of SLA show only modest levels of genetic variation.

The main premise of this study was to determine whether or not the seeds from the UPEI seed bank could be used to serve as seed stock to re-establish SLA populations through in situ transplant and (or) seeding strategies. The primary objective was to determine the best methods for establishing self-sustaining populations of SLA. Our central hypothesis was that SLA would benefit from simulation of natural disturbance to the marsh substrate such as the creation of potholes and the removal of competing vegetation. A secondary objective was to gain a better understanding of the abiotic factors that may be influencing the species at the test sites on PEI.

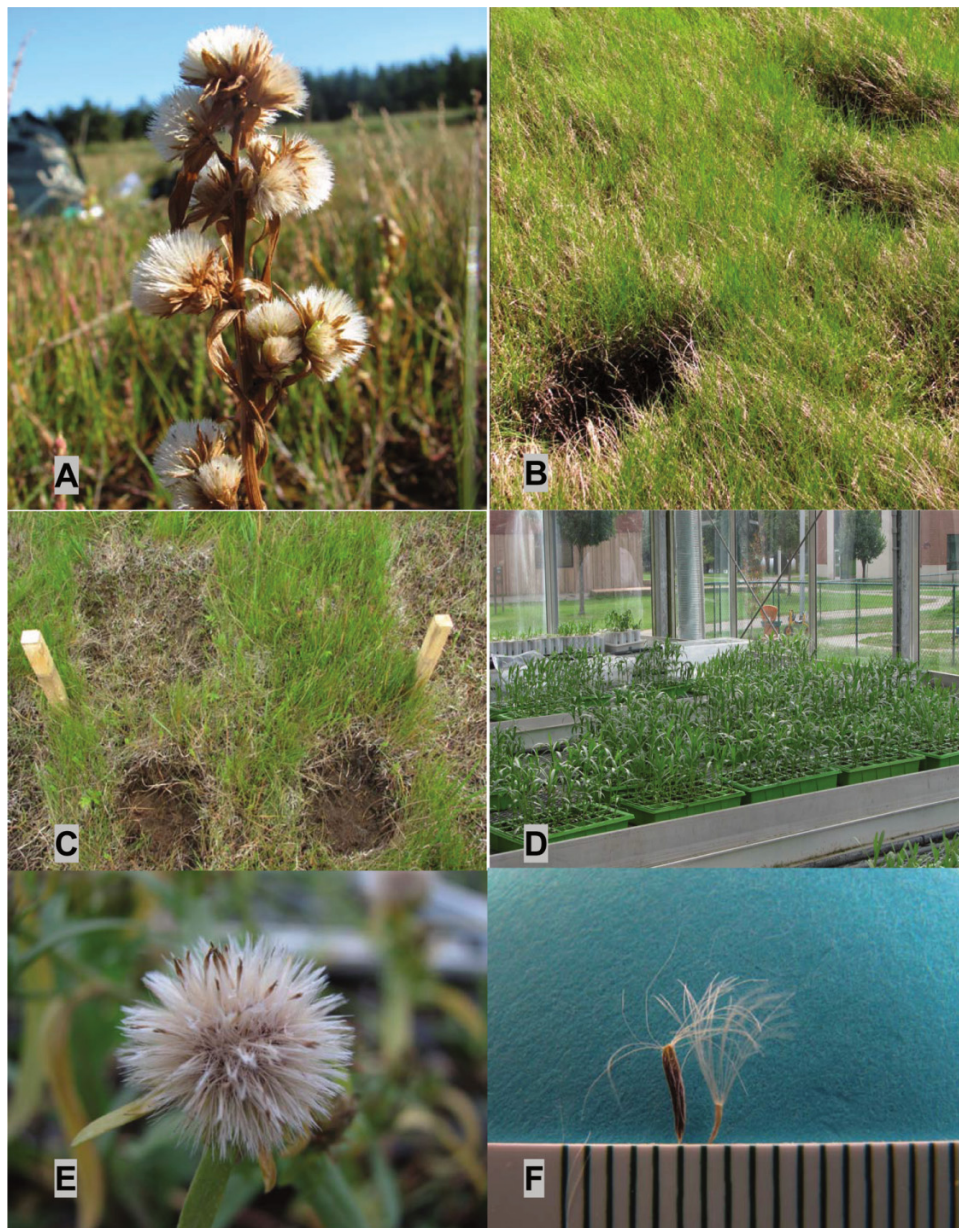
Materials and methods

Two-year-old achenes of SLA produced in the UPEI greenhouse in 2006 were used in this study. This particular seed stock originated from a collection of inflorescences that was gathered in 1999 from the East Marsh location of Blooming Point on PEI. The seedlings were grown in a greenhouse setting under light and temperature levels commensurate with seasonal conditions and were transplanted to four predetermined sites within PEINP. Ex situ seed germination and plantlet production was synchronized with the phenology of plants in their natural setting.

Site selection and preparation

Site selection for this project was based on historical records, as well as physical, biological, and logistical suitability of candidate areas. In total, four sites (Blooming Point East Marsh A and B, Blooming Point Dune Slack, and Robinson's Island) were selected to conduct the experiments (Houle and Haber 1990). The first three sites were historical (reintroduction) sites, while the fourth was a nonhistorical (introduction) site (Fig. 1). At the end of July, the selected sites were manipulated to receive transplants. Two grids (1 m × 3 m) were set up at each site in 2008 and 2009. A randomized block design was used to establish four plots within each grid for each of the following

Fig. 2. *Symphotrichum laurentianum* habitat photos and experimental setup. (A) Mature plant in situ. Note the distinct white pappus on the florets within the inflorescences. 0.25X. (B) Natural “potholes” found within the substrate of the East Marsh at Blooming Point. Note the reduction of competing vegetation within the potholes. (C) Test plots within grid at Blooming Point]: (top left) vegetation removed; (top right) untouched; (bottom left and right) potholes. (D) Gulf of St. Lawrence aster (SLA) plants in greenhouse, prior to transplantation (approximately 2 months of age). (E) Mature SLA inflorescence growing in the University of Prince Edward Island (UPEI) greenhouse, prior to collection and planting of the fall seeding plots, 2008. 1X. (F) Image of filled SLA achene (left) and unfilled achene (right). For this study, counts of filled achenes were used to estimate potential viability. Each division on the scale represents 1 mm.



three transplanting strategies: undisturbed marsh vegetation (control), mowed marsh vegetation, and artificially simulated potholes (Fig. 2C). By the end of the second field season, there were 64 plots for each of the three treatment groups tested.

Seedling and transplant preparation

Seedling production (*ex situ*)

In 2008, several hundred seeds of SLA were germinated using a standard potting soil (PG Mix, professional growing mix) medium in a growth chamber. The chamber was set to a 14 h light – 10 h dark photoperiod with a thermal regime of 25 °C day – 15 °C night and 80% humidity. After 2 weeks, seedlings of similar size (cotyledon stage) were transplanted into individual 25 cm³ cells filled

with the same mix of soil. These seedlings were moved to the greenhouse to continue their development for approximately 2 months (Fig. 2D). The greenhouse was set to maintain temperature levels between 20 and 30 °C throughout the day and 15 and 21 °C throughout the night. The natural photoperiod of the season was used (approximately 14 h of light). The plants were watered manually on a daily basis and monitored for any disruption (e.g., aphid infestation).

Hardening off

Over the course of 2 weeks prior to transplantation, the SLA seedlings were slowly acclimated to natural weather conditions. This was accomplished by increasing the time, by approximately

2 h, that the plantlets spent outside of the greenhouse each day. By the end of the hardening-off period, the seedlings were acclimated to strong sunlight, cool nights, and less-frequent watering.

Transplantation and monitoring

At approximately 2 months postgermination, when SLA plants had achieved a vigorous growth, but had not yet initiated flowering, 10 plantlets were transplanted to each of the established plots in calm weather conditions during neap tides. These conditions were chosen to reduce the probability that bad weather or high tides would displace and (or) disrupt the acclimation of seedlings or the transplant process. Each transplant was watered with fresh water immediately after planting to further assist with acclimation to higher in situ salinity levels by making the transition more gradual.

Inflorescence planting

In a natural setting, mature SLA seed heads often remain intact and become embedded in the soil as a single unit. Therefore, plots were created on the periphery of the 2008 transplant grids to test the success of germination from whole inflorescences planted at two different times of the year. Fifteen mature inflorescences were planted in 0.5 m² plots in the fall of 2008 and in the spring of 2009. The plots were manipulated slightly to facilitate the embedding of the seeds in the soil; as a result, approximately 50% of the existing vegetation was pulled out of the substrate prior to seeding.

The inflorescences used were collected from the 2008 greenhouse-grown plants at maturity (at the beginning of dispersal) (Fig. 2E). The inflorescences used for the fall plots were planted immediately after they were collected. Inflorescences for the spring plots were stored in a refrigerator in the UPEI seed bank over the 2008 and 2009 winter months prior to planting at the end of May 2009. These seeded plots were monitored to determine if successful germination and survival of plants took place during the 2009 season.

Developmental variables monitored

Survivorship to seed set

Monitoring of seedling survival commenced 1 week after transplantation and continued biweekly until seed dispersal occurred (late October – early November). Survival was considered to be achieved if the plants remained green, intact, and continued developing on site. The mean number of inflorescences per plant per treatment was also noted.

Seed production and potential

A proportion (10%) of the inflorescences was collected from each plot once the achenes had matured. In the laboratory, the total number of achenes produced within these inflorescences was recorded. Visual inspection was used to classify the achenes as being either filled or unfilled to determine percentage values for each category. Filled achenes contain an embryo and are noticeably wider and longer (2.5–3.5 mm) than unfilled achenes. Filled achenes are typically reddish to dark brown in colour, whereas unfilled achenes are light-brown to tan coloured (Jenkins 2008) (Fig. 2F). Once counted, the achenes were returned (spread on) to their respective grids in the field to avoid reduction in the potential seed bank for the 2009 season. Although the viability of the seeds was not tested, the visual inspection of the achenes was used as a method of gaining quantitative estimates of potential viability.

Germination surveys

In early June of 2009, surveys were conducted at the management sites to look for second generation plants. Each plot was thoroughly searched, in addition to 1 m around the perimeter of each grid.

Abiotic conditions monitored

Weather conditions

Data collected from a Vantage Pro 2 Weather Station (Davis Instruments, Hayward, California, USA) set up approximately 1 km south of the Blooming Point reintroduction sites were used to monitor environmental conditions for the duration of the experiment. Air temperature, rainfall, wind speed, and wind direction were recorded hourly between May and October for each study year.

Wrack and eelgrass monitoring and management

The presence and (or) absence of wrack build up, both within and around the grids, was noted on a biweekly basis. One of the two grids at each of the experimental sites for each of the 2 years of study was managed for eelgrass and (or) wrack buildup. This was accomplished by removing any buildup of wrack within the grid on a biweekly basis.

Ground temperature, moisture, and soil salinity

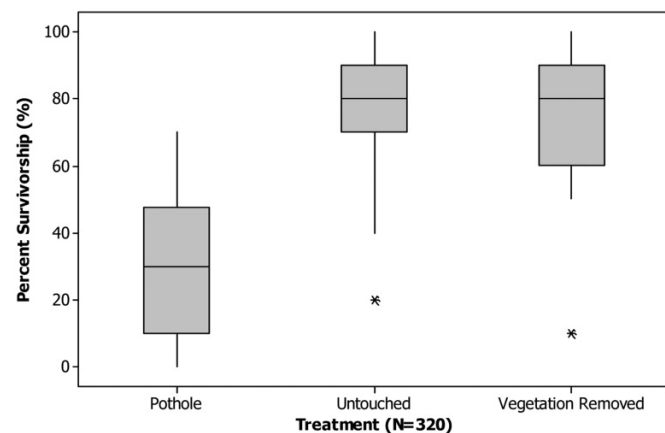
These abiotic conditions were monitored in context at the testing sites to determine if conditions were relatively uniform across all sites. Temperature loggers (Hobo Tidbit version 2 Temp Logger Hoskin Scientific Ltd., Burlington, Ontario, Canada) were inserted 7.62 cm into the soil at each site at the beginning of the summer and remained in the soil for the duration of the plants' development to record ground temperature on a daily basis. A standard Westminster soil moisture meter, with a scale of 0–10 (0 representing dry and 10 representing wet), was used biweekly to record relative moisture levels within the grids.

Throughout the germination season, soil samples were collected during high tide and in close proximity to each of the two grids to estimate soil salinity at each of the four management sites. The three collection dates corresponded to late May – early June, mid-June, and end of June. Samples averaged 250 mL in volume and were dried at 40 °C for 48 h. Samples were allowed to reach room temperature and were sifted through a 2 mm sieve. Soil salinity was determined by measuring the electrical conductivity (EC) of a mixture of 1 part soil to 5 parts distilled water (10 g : 50 mL). The soil and water were continuously mixed on a mechanical rotary shaker at 80 osc/min for 1 h before the EC was tested. After allowing the solution to settle for 1 min, a Thermo Fisher Scientific Symphony Conductivity Meter (Hoskin Scientific Ltd.) was used to gather an indirect measure of the total soluble salt concentration in the soil solution (measurements were taken above the settled soil). The unit of measurement was decisiemens per metre (dS/m), which was then converted to grams per litre (g/L) given the soil solution ratio used. Three separate solutions were created for each of the two samples taken at each site. These results were then averaged to provide an estimated value for the site on each sample date. This methodology was provided by the PEI Department of Agriculture, Fisheries and Aquaculture and is their standard method for testing soil salinity.

Analysis

After log transforming the survivorship data, the assumptions for parametric analysis of variance testing were met. For comparison of final survivorship among each year tested and the three treatment methods, a factorial analysis of variance (two-way ANOVA) testing was used. Post hoc analysis (the Tukey test) was carried out for the three treatment methods to compare results for each year. Analysis of covariance (ANCOVA) was used to determine site effects on the treatment results. All of these analyses were conducted with Statistica statistical software (Statsoft, version 9 StatSoft Inc., Tulsa, Oklahoma, USA). Additional analysis consisted of conducting Mann–Whitney *U* tests to compare the mean percentage of filled seeds for each treatment in 2008.

Fig. 3. Box plots of the distribution of survivorship data for the 2008 Gulf of St. Lawrence aster (SLA) transplants of each treatment group: pothole, untouched, and vegetation removed. The center line represents the median and the asterisks represent the outliers.



Results

Final survivorship to seed set

Site had no significant effect on the treatment results (ANCOVA; $p = 0.06$); therefore, all of the sites were pooled for further analysis. Year did have a significant effect (two-way ANOVA; $p = 0.0004$) on the survivorship of the plants; consequently, each year was analyzed separately. Interestingly, for each year, the only significant differences were between the pothole and vegetation-removed groups (Tukey; $p = 0.00002$ (2008) and $p = 0.00002$ (2009)) and the pothole and untouched groups (Tukey; $p = 0.00002$ (2008) and $p = 0.00002$ (2009)). These differences were more prominent in 2009 than in 2008 (Figs. 3 and 4). The overall survivorship of the transplants was 63% in 2008 and 47.7% in 2009. Over the 2 years of the experiment, the pooled overall survivorship was 52.8%.

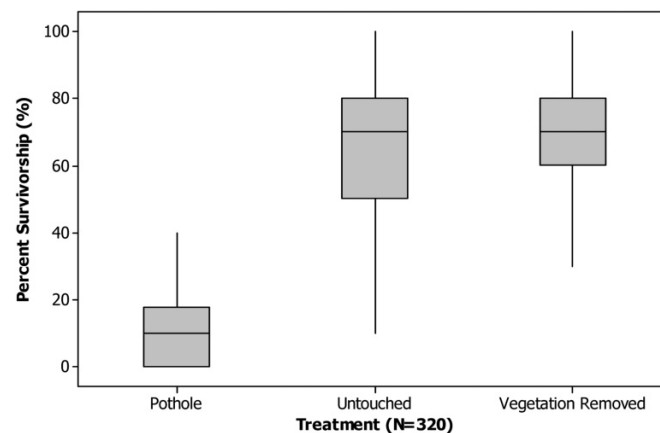
Mean number of inflorescences per plant, per treatment

Throughout the month of October, transplants were monitored for survivorship and seed dispersal. The mean number of inflorescences was lower on the plants in the pothole treatment compared to the other two treatments. The pothole plants averaged 3.6 ± 2.3 inflorescences per plant, the vegetation-removed plants averaged 6.1 ± 4.1 , and the control or untouched plants averaged 6.8 ± 4.5 (means $\pm$ SDs).

Seed production and potential

In October of 2008, a total of 271 inflorescences were collected from the grids. These represented approximately 10% of the total number of inflorescences located in each plot at the time of plant maturity. There were no significant differences between the treatment groups tested (for all three treatment comparisons, Mann-Whitney U tests; $p > 0.05$). Twenty-five inflorescences were examined from the pothole-treatment group, with a mean of $70.19\% \pm 18\%$ filled seeds. For the untouched and vegetation removed treatment groups, 122 and 124 inflorescences were examined, respectively. The plants in the vegetation removed treatment group had a $70.23\% \pm 21\%$ filled seeds in their inflorescences, and plants in the untouched treatment group had $67.18\% \pm 19\%$ filled seeds in their inflorescences (means $\pm$ SDs). It was also noted that the majority of the dispersed seeds seemed to remain within the immediate area surrounding the maternal plants, a phenomenon known as "seed shadowing" (Lacroix et al. 2007). This spatial distribution was useful in detecting second-generation SLAs derived from the transplants.

Fig. 4. Box plots of the distribution of survivorship data for the 2009 Gulf of St. Lawrence aster (SLA) transplants of each treatment group: pothole, untouched, and vegetation removed. The center line represents the median and the asterisks represent the outliers.



In situ germination surveys

These surveys led to the positive identification of over 500 SLA seedlings found within and around the 2008 management grids. Owing to limited time and human resources, a simple survey count was conducted to gather these data. However, it was noted that at each site the seedlings were found growing within areas where vegetation reduction had taken place, including the edges of the potholes and where the substrate had been manipulated in the untouched and vegetation-removed plots to insert the transplants. On a treatment level, the majority of the seedlings were found growing in clusters in the fall seeding plots (Fig. 5). Unfortunately, the seedlings died 1 week later due to extreme weather conditions.

Weather challenges

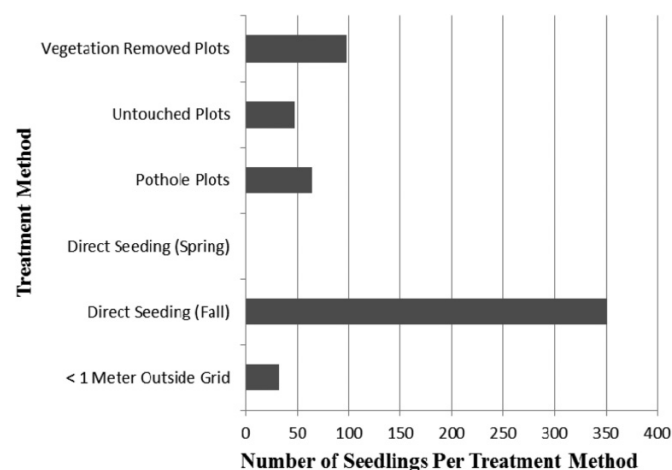
Over the course of the 2008 and 2009 seasons, the experimental plants experienced extreme weather conditions. In 2008, transplanting at the Dune Slack site was delayed due to heavy flooding of the marsh. Throughout this study, both the East Marsh A and B sites, located adjacent to the Dune Slack site, dried up much faster than the Dune Slack site, which retained water for up to 4 weeks longer. The flooding is believed to be the result of heavy rainfall that took place throughout the month of August. According to the weather office located at the Charlottetown Airport, August 2008 was the wettest month ever recorded on PEI. The month's total precipitation recorded at the Blooming Point weather station was 191.6 mm.

Within 1 week of transplantation, which took place in August, all of the 2008 experimental plots were inundated by flooding due to the high precipitation levels. Despite these conditions, the plants survived even though many of them, especially plants in the pothole treatment, were completely submerged for several days. The SLA plants apparently aborted the development of submerged buds but were able to maintain the development of the terminal and lateral buds that were above the water and were exposed to air and sunlight.

Another major rain event occurred during the germination period in June 2009, when 97.2 mm of rain fell within 24 h. This rainfall event may have been responsible for the loss of germinating SLA seedlings resulting from the 2008 SLA transplants.

Weather also affected the transplanting in 2009. Within 2 weeks of planting, the 2009 transplants were exposed to high winds and rains from the remnants of two Atlantic hurricanes, Hurricane Bill and Hurricane Danny. The first storm, on 23 August, had wind speed up to 64.4 km/h and total precipitation of 18.2 mm at Blooming Point. One week later, the second storm had wind speeds up to

Fig. 5. Gulf of St. Lawrence aster (SLA) second-generation germination counts for each treatment method tested. Germination surveys took place in June of 2009.



51.5 km/h and 108.4 mm of rainfall. The majority of the transplants survived these storms; only two plants were uprooted and displaced from their plots.

Wrack and eelgrass monitoring and management

Throughout the development of the SLA transplants, little to no eelgrass removal was necessary at any site. Although eelgrass would build up along the beach in thick mats, very little washed up into the sites and created issues for the transplants prior to seed dispersal; therefore, no conclusions could be drawn from the eelgrass management efforts on the survivorship of the relatively mature transplanted SLAs. However, at the time of germination in 2009, there were plots within some grids that were buried under mats of dead eelgrass up to 10 cm thick. Despite management actions to remove the built-up eelgrass, grid comparisons were not possible due to the death of all of the germinating seedlings.

Ground temperature, moisture, and soil salinity

The following abiotic factors were relatively uniform across all testing sites. The mean ground temperature at the time of transplantation in 2008 was 18.9 ± 1.4 °C, while 2009 temperatures were higher at 23.3 ± 5.2 °C. The mean ground temperature during SLA germination in 2009 was 15.29 °C ± 4.0 (means $\pm$ SDs). Although some sites were visibly wetter or drier than others at times, each moisture measurement taken indicated more or less uniform conditions throughout all sites. The substrate remained wet and always rated 10 on the meter's scale.

In all samples and at all sites, the soil salinity never rose above 8 g/L. The historical sites displayed low levels of soil salinity each time they were tested, with the highest levels being 1.08 g/L at the East Marsh A site. On the other hand, Robinson's Island, the only nonhistoric site tested, did reach higher salinity levels, with a maximum of 7.7 g/L and a minimum of 6.4 g/L.

Discussion

Although it was hypothesized that removing vegetation and simulating natural potholes would aid in population establishment, our survival results indicate otherwise. The survivorship results indicate that transplants did not benefit from the removal of vegetation to lower competition levels, and that simulating potholes had a detrimental effect on populations. This was presumably due in large part to the frequent flooding and consistent waterlogging of the pothole plots.

Seed dispersal refers to the ability of seeds to spread through space and time within occupied sites or to unoccupied sites (Baskin and Baskin 1998). The dispersal potential of a plant can

affect both the regional and local distribution of that species (Primack and Miao 1992). Achenes of the SLA possess a pappus, or plume-like structure, that slows their descent and may aid in transporting them farther away from the mother plant. Lacroix et al. (2007) assessed SLA dispersal potential and found that in situ achenes travelled a maximum of 23 cm from their point of release. Travel was impeded when the achenes came into contact with the surrounding vegetation. They also noted that seedlings were generally found in the immediate area surrounding the previous years' plants. Based on the seed shadow of the plants studied, most achenes travelled a range of 0.15–0.20 m.

The seed shadow phenomenon (Willson 1993) was also witnessed during both seasons of this study. The dispersal trends observed were very similar to those highlighted in Lacroix et al. (2007) with the spatial distribution of the majority of the seeds remaining close to the source. For this reason, positive identification of second-generation SLA plantlets within and around the experimental plots was assumed to have derived from the adjoining treatment plots themselves. The majority of the seedlings located during the germination surveys were found growing in clusters in the fall seeding plots.

These results indicate that the seeding methodology that was utilized in the fall served the purpose of initiating SLA plant populations in situ. Although the survivorship data indicate that there was not a significant difference in untouched and vegetation-removed treatments, higher germination was noted to take place in the areas where vegetation underwent some form of removal, including the upper edges of the potholes and areas within the plots where vegetation was disturbed to facilitate the insertion of the transplants the previous year. This suggests that these manipulations may affect the overall reproductive potential of the transplants and (or) seeds by facilitating second-generation plant establishment. The reduction in competition from other plants is believed to play an important role in SLA's establishment as this species tends to colonize reasonably bare substrates and requires disturbance to offset competition (Gilbert et al. 1999; Houle et al. 2001; Houle and Valéry 2003).

Throughout both field seasons, plants in the pothole treatment remained underwater and waterlogged longer than plants in the other treatments. These pothole plants had the lowest survival rates. Houle and Haber (1990) suggested that intra- and inter-annual water level fluctuations affect the annual variations in population density of this species. In 1999, Houle and Belleau (2000) studied the effects of drought and waterlogging on the development and performance of the SLA. They found that drought stress can negatively affect flower-head production, but that waterlogging had no effect on plant growth or biomass allocation. Although our study was unable to provide additional in situ knowledge on drought stress due to the high amounts of precipitation, extreme waterlogging and prolonged flooding was a common factor experienced by the transplants, and may have contributed to the low survivorship rates of plants in the potholes. It is possible that our results differed due to the duration of waterlogging and extent to which our plants were immersed in water.

There was no difference when comparing the estimated potential viability of filled and unfilled achenes across the three treatment groups. All groups produced relatively equal mean percentages of filled achenes (approximately 70%). These means are consistent with the data collected by Lacroix et al. (2007). They found that inflorescences collected from the Dune Slack and East Marsh B sites ($n = 47$) in 2002 produced 64.9% filled achenes. There are, however, acknowledged limitations to the collection of data for the 2008–2009 study. The data that were collected cannot be considered completely unbiased. Owing to the need to collect inflorescences that looked as though they had not lost many seeds to dispersal and achenes that were mature, it is possible that the inflorescences chosen represented a biased sample of productive

inflorescences. However, given the consistent nature of our results, this concern may be minimal. For this reason, inflorescences were only inspected after the 2008 season.

Unfortunately, we were not able to determine whether eelgrass wrack deposition had any effect on germination or survivorship of SLA. However, it is still possible that eelgrass deposition is a predominant factor that negatively affected the SLA seed bank and germination success at certain sites and could be tested further. Although eelgrass seemed to cover the plant habitat only during germination in the 2008 and 2009 seasons, eelgrass removal would be most beneficial if conducted prior to and throughout the germination season after the accumulation of dead eelgrass over the fall, winter, and spring months (Hemminga and Duarte 2000). These accumulations of wrack have the ability to bury the seeds in the seed bank and inhibit their germination. It is also possible for mats of eelgrass located within the marsh at the time of SLA germination to be moved by spring floods and winds, and consequently have the same effect. As Reynolds and Houle (2003) suggest, the relative importance of this structuring variable changes with the physical characteristics of the site. Ongoing monitoring and experimental management of this factor may be worthwhile.

According to Houle et al. (2001), SLA plants are more susceptible to high salinity during germination and early seedling growth than in later stages of development. They demonstrated that germination is reduced once salt concentrations reach 10 g/L and is completely inhibited once concentrations reach 20 g/L. The levels detected in the soil salinity tests conducted throughout the germination season in 2009 were lower than the cautionary or limiting levels listed by Houle et al. (2001). Our tests revealed levels to be quite uniform across the testing sites, with maximum salinity levels reaching 7.7 g/L. Although there is no conclusive or quantitative evidence to prove that the seed bank at these sites was affected by salinity levels, the 2009 germination counts suggest that levels were at least within an acceptable range to produce seedlings.

Recent research conducted by Heard and Ancheta (2010) proposed that the effect of temperature on SLA germination is more limiting than the effect of salinity. They state that delays in germination resulting from exposure to low soil temperatures could set up strong size asymmetries between SLA seedlings and surrounding vegetation, resulting in a suppression of SLA population growth due to shading. Interestingly, the majority of the temperatures that they tested (16, 23, and 30 °C) were above the mean ground temperature recorded in situ for this experiment (15.29 ± 4.0 °C). They found that after 16 days, germination was greatest at their highest temperature tested. Therefore, the results from the 2008–2009 study suggest that ground temperature may not be optimal for SLA establishment throughout the germination season at the sites tested within PEINP.

Although there are many factors to consider when attempting to establish populations of SLA, this study concludes that recovery efforts based on the methodologies listed may be possible. As the strategic direction for this species is considered through the proposed recovery strategy developed by Environment Canada (2012), it is hopeful that these data will provide actions and guidelines for future recovery efforts in identified target sites across the species region. Discerning which methods would be most successful at each recovery site would be more precise and would aid in the goal of creating a self-sustainable population. Therefore, adequate information, such as salinity levels, gathered on the potential sites that will be used for future recovery purposes would be beneficial.

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References

- Allain, M. 2007. Seed bank management and transplanting success of the Gulf of St. Lawrence Aster (*Symphyotrichum laurentianum*). Unpublished Honours thesis, Department of Biology, University of Prince Edward Island, Charlottetown, Canada.
- Baskin, C.C., and Baskin, J.M. 1998. Seeds: ecology, biogeography, and evolution of dormancy and germination. Academic Press, San Diego.
- Committee on the Status of Endangered Wildlife in Canada (COSEWIC). 2004. Species listing, terms and risk categories. Available from <http://www.cosewic.gc.ca> [accessed May 2008].
- Environment Canada. 2012. Recovery Strategy for the Gulf of St. Lawrence Aster (*Symphyotrichum laurentianum*) in Canada (Proposed). Species at Risk Act Recovery Strategy Series, Environment Canada, Ottawa.
- Fernald, M.L. 1914. Some annual halophytic asters of the Maritime Provinces. *Rhodora*, **16**: 57–61.
- Gilbert, H., Labrecque, J., and Gagnon, J. 1999. La situation de l'aster du Saint-Laurent (*Aster laurentianus*, syn.: *Symphyotrichum laurentianum*) au Canada. Gouvernement du Québec, ministère de l'Environnement, Direction de la conservation et du patrimoine écologique, Québec.
- Given, D.R. 1994. Principles and practices of plant conservation. Timber Press, Oregon.
- Harrison, S., and Bruna, E. 1999. Habitat fragmentation and large-scale conservation: what do we know for sure? *Ecography*, **22**: 225–232. doi:10.1111/j.1600-0587.1999.tb00496.x.
- Heard, S., and Ancheta, J. 2010. Effects of salinity and temperature on germination of the threatened Gulf of St. Lawrence Aster, *Symphyotrichum laurentianum* Fernald (Nesom). *Plant Species Biology*, **26**: 158–162.
- Heard, S.B., Jesson, L.K., and Tulk, K. 2009. Population genetic structure of the Gulf of St. Lawrence aster, *Symphyotrichum laurentianum* (Asteraceae), a threatened coastal endemic. *Botany*, **87**(11): 1089–1095. doi:10.1139/B09-068.
- Hemminga, M., and Duarte, C. 2000. Seagrass ecology. Cambridge University Press, New York.
- Houle, F. 1988. Étude biosystématique de la section *Conyzopsis* du genre *Aster* (Asteraceae). Thèse présentée à la faculté des études supérieures en vue de l'obtention du grade de Ph.D en sciences biologiques, Université de Montréal.
- Houle, F., and Haber, E. 1990. Status of the Gulf of St. Lawrence Aster, *Aster laurentianus* (Asteraceae), in Canada. *Can. Field-Nat.* **104**: 455–459.
- Houle, G., and Belleau, A. 2000. The effects of drought and waterlogging conditions on the performance of an endemic annual plant, *Aster laurentianus*. *Can. J. Bot.* **78**(1): 40–46. doi:10.1139/b99-158.
- Houle, G., and Valéry, S. 2003. A mixed strategy in the annual endemic *Aster laurentianus* (Asteraceae) – a stress-tolerant, yet opportunistic species. *Am. J. Bot.* **90**: 278–283. doi:10.3732/ajb.90.2.278. PMID:21659119.
- Houle, G., Morel, L., Reynolds, C.E., and Siegel, J. 2001. The effect of salinity on different developmental stages of an endemic annual plant, *Aster laurentianus* (Asteraceae). *Am. J. Bot.* **88**: 62–67. doi:10.2307/2657127. PMID:11159127.
- Jenkins, E. 2008. Conservation and Pollination of *Symphyotrichum laurentianum*, the Gulf of St. Lawrence Aster. Unpublished Honours thesis, Department of Biology, University of Prince Edward Island, Charlottetown, Canada.
- Kemp, J.F., and Lacroix, C.R. 2004. Estimation of seed bank and seed viability of the Gulf of Saint Lawrence Aster, *Symphyotrichum laurentianum* (Fernald) Nesom. *Can. Field-Nat.* **118**: 105–110.
- Lacroix, C.R., Steeves, R., and Kemp, J.F. 2007. Floral development, fruit set, and dispersal of the Gulf of St. Lawrence aster (*Symphyotrichum laurentianum*) (Fernald) Nesom. *Can. J. Bot.* **85**(3): 331–341. doi:10.1139/B07-011.
- Primack, R.B., and Miao, S.L. 1992. Dispersal can limit local plant distribution. *Conserv. Biol.* **6**: 513–519. doi:10.1046/j.1523-1739.1992.06040513.x. PMID:10.1046/j.1523-1739.1992.06040513.x.
- Reynolds, C.E., and Houle, G. 2001. Mantel and partial Mantel tests suggest some factors that may control the local distribution of *Aster laurentianus* at les Iles-de-la-Madeleine, Quebec. *Plant Ecol.* **164**: 19–27.
- Steeves, R. 2005. Seed dispersal potential, pre-dispersal seed predation, and population estimates of the Gulf of St. Lawrence Aster (*Symphyotrichum laurentianum*) in PEI National Park. Unpublished Honours thesis, Department of Biology, University of Prince Edward Island, Charlottetown, Canada.
- Stewart, S.E., and Lacroix, C.R. 2001. Germination potential, updated population surveys and floral, seed and seedling morphology of *Symphyotrichum laurentianum*, the Gulf of St. Lawrence Aster, in the Prince Edward Island National Park. *Can. Field-Nat.* **115**: 287–295.
- Willson, M.F. 1993. Dispersal mode, seed shadows, and colonization patterns. *Plant Ecol.* **108**: 261–280. doi:10.1007/BF00052229.
- Wilson, E.O. 1992. The diversity of life. Belknap Press of Harvard University Press, Cambridge, Mass.